



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
FULL SYLLABUS
12-07-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-II (ALL STAR BATCH)]

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

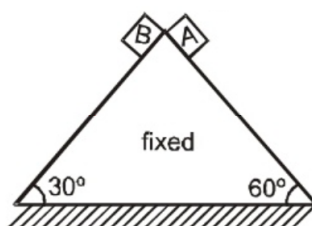
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Small blocks A and B are simultaneously released from apex of a smooth wedge as shown in figure. Select correct alternative(s) :



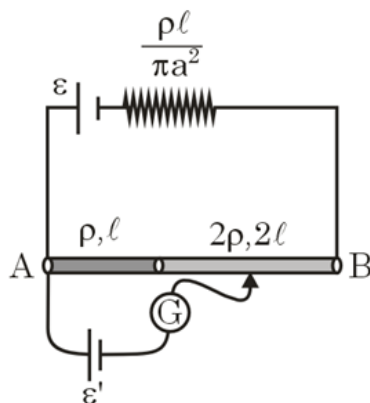
- (A) Relative acceleration of block B with respect to block A is zero.
- (B) Magnitude of relative acceleration of block B with respect to block A is g initially
- (C) Speed of block A and B will be same at the bottom of inclined plane
- (D) The time taken by A and B to reach the bottom of inclined plane will be same.

Space for Rough Work

2. Two particles A and B having mass m and $2m$ are placed 'd' distance apart. Particle A is thrown with speed v . Such that it escapes gravitational pull of other particle B. What should be minimum value of v ?

(A) $\sqrt{\frac{2GM}{d}}$ (B) $\sqrt{\frac{6GM}{d}}$ (C) $\sqrt{\frac{GM}{d}}$ (D) $\sqrt{\frac{2GM}{3d}}$

3. In a potentiometer primary cell has internal resistance of $\frac{\rho\ell}{\pi a^2}$ as shown in figure. the potentiometer wire is combination of two wire connected in series as shown in figure. The common cross section radius of both wire used for potentiometer is 'a'.



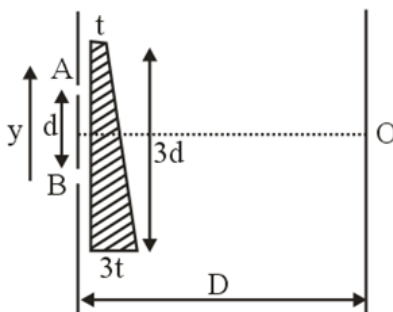
- (A) Range of potentiometer is $\frac{5\varepsilon}{6}$
 (B) Range of potentiometer is $\frac{2\varepsilon}{3}$
 (C) If null point is at mid point of longer wire then $\varepsilon' = \frac{\varepsilon}{2}$
 (D) If null point is at mid point of longer wire then $\varepsilon' = \frac{\varepsilon}{4}$

Space for Rough Work

4. X-ray from a tube with a target A of atomic number Z shows strong K_α lines for target A and two weak K_α lines for impurities. The wavelength of K_α lines is λ_0 for target A and λ_1 and λ_2 for two impurities respectively.
- $\frac{\lambda_0}{\lambda_1} = 4$ and $\frac{\lambda_0}{\lambda_2} = \frac{1}{4}$. The screening constant of K_α lines is unity. Select the correct alternative(s) :
- (A) The atomic number of first impurity is $2Z - 1$
- (B) The atomic number of first impurity is $2Z + 1$
- (C) The atomic number of second impurity is $\frac{Z + 1}{2}$
- (D) The atomic number of second impurity is $\frac{Z}{2} + 1$

Space for Rough Work

5. In a standard YDSE two slits are covered by a wedge shape film of refractive index μ as shown in the figure. The thickness of film varies linearly from t to $3t$ from one end as shown in the figure. The distance between the two slits is d . The screen is at distance D from slits. White light is used in experiment. Choose the correct option(s) : ($d \ll D$)



Select correct alternative(s)

- (A) Central maxima is formed at distance $\frac{2t(\mu - 1)D}{3d}$ below O
- (B) If the film is displaced $\frac{d}{2}$ along +y axis the centre maxima will shift away from O
- (C) If the film is displaced $\frac{d}{2}$ along -y-axis the centre maxima will shift away from O
- (D) If the film is displaced $\frac{d}{2}$ along +y axis the fringe width will remain same
6. A concave spherical mirror's surface is defined by $x^2 + y^2 + z^2 = 16$ with pole at point $(4, 0, 0)$. A ray of light parallel to x axis is incident on mirror at point $(\sqrt{12}, \sqrt{3}, 1)$. Choose the correct option.
- (A) Reflected ray cuts principal axis at point $\left(\frac{4}{\sqrt{3}}, 0, 0\right)$.
- (B) Reflected ray cuts, principal axis at point $\left(\frac{2}{\sqrt{3}}, 0, 0\right)$
- (C) Deviation suffered by ray during reflection is 120°
- (D) Deviation suffered by ray during reflection is $180 - 2\sin^{-1} \frac{1}{\sqrt{3}}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

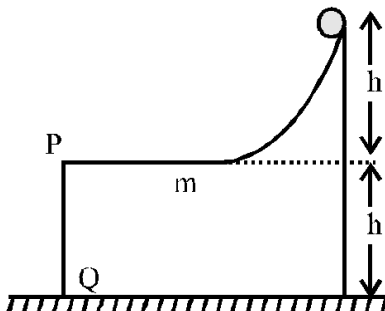
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A small ball (uniform solid sphere) of mass m is released from the top of wedge of the same mass m . The wedge is free to move on a smooth horizontal surface. The ball rolls without sliding on the wedge. The required height of the wedge are as mentioned in the figure.

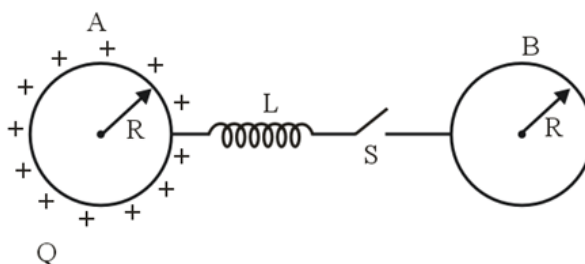


7. The speed of the wedge when the ball is just going to leave the wedge at point 'P' of the wedge is
- (A) $\sqrt{\frac{5gh}{9}}$ (B) $\sqrt{gh}$ (C) $\sqrt{\frac{5gh}{6}}$ (D) None of these
8. The total kinetic energy of the ball just before it falls on the ground
- (A) $2 mgh$ (B) mgh (C) $\frac{31}{18} mgh$ (D) None of these

Space for Rough Work

Paragraph for Questions 9 and 10

Two conducting sphere of radius R are placed at a large distance from each other. They are connected by a coil of inductance L , as shown in the figure. Neglect the resistance of the coil. The sphere A is given a charge Q and the switch 'S' is closed at time $t = 0$.



9. Find charge on sphere B as a function of time.

(A) $\frac{Q}{3} [1 - \cos \sqrt{\frac{2K}{LR}} t]$ (B) $\frac{Q}{2} [1 - \cos \sqrt{\frac{3K}{LR}} t]$
 (C) $\frac{Q}{2} [1 - \cos \sqrt{\frac{2K}{LR}} t]$ (D) $\frac{Q}{3} [1 - \cos \sqrt{\frac{3K}{LR}} t]$

10. At what time charge on B is $\frac{Q}{2}$?

(A) $\frac{\pi}{2} \sqrt{\frac{LR}{3K}}$ (B) $\frac{\pi}{2} \sqrt{\frac{LR}{2K}}$ (C) $\frac{\pi}{4} \sqrt{\frac{LR}{2K}}$ (D) $\frac{\pi}{4} \sqrt{\frac{LR}{3K}}$

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

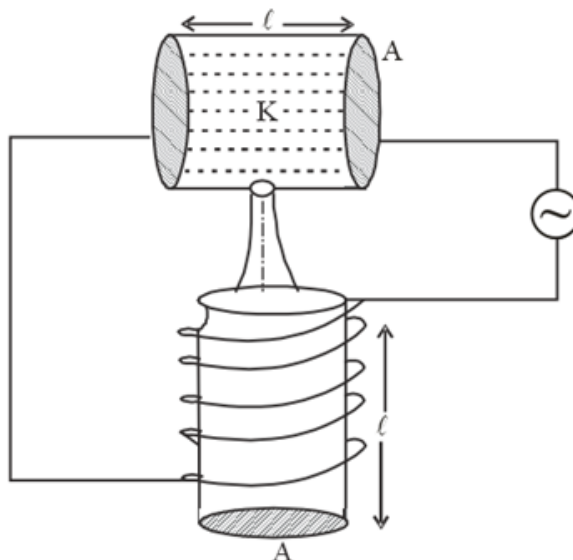
A planet is revolving around a sun in elliptical orbit whose equation is given by $7x^2 + 16y^2 = 112$

1. What is ratio of maximum to minimum speed of planet?
2. By what percentage speed of planet at Aphelion must be increased so than planet starts orbiting is circular path?

Space for Rough Work

Paragraph for Questions 3 and 4

Figure shows a system of inductor and parallel plate capacitor made of 2 parallel circular plates of area A and filled with dielectric liquid of dielectric constant K as shown. A small leak develops in capacitor and liquid starts to fill the inductor of same dimensions having n turns / unit length.



3. Find the ratio of magnitude of initial to final reactance of circuit after liquid fills the inductor completely
 Given : $\omega^2 A^2 n^2 = c^2$
 $\omega \rightarrow$ angular frequency of AC
 $c \rightarrow$ speed of light
 and $\mu_r \rightarrow$ relative permeability of liquid
 $K = 2, \mu_r = 3$
4. If the value of resistance R that should be added in series to the circuit at an instant when exactly half of the liquid has flown from capacitive cylinder to inductive cylinder for which power factor of the circuit at that instant becomes $\frac{1}{\sqrt{2}}$ comes out to be equal to $R = \frac{\alpha l}{3\epsilon_0 A \omega}$ then find the value of α .

Space for Rough Work

Paragraph for Questions 5 and 6

The driver of a car travelling at 100 Km/hr towards a cliff briefly sounds the horn. Exactly one second later he hears the echo and notes that its frequency is 840 Hz. (speed of sound 330 m/s)

5. How far from the cliff was the car when the driver sounded the horn?
6. What is the actual frequency of the car horn?

Space for Rough Work

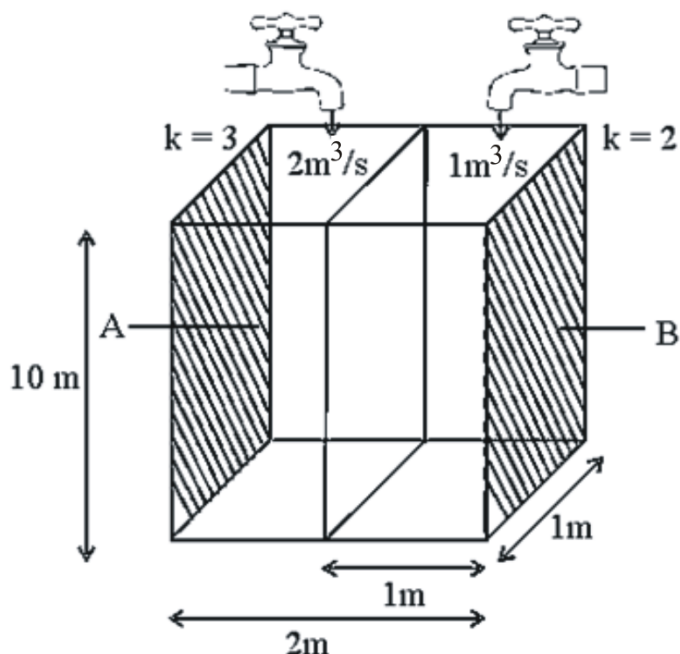
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

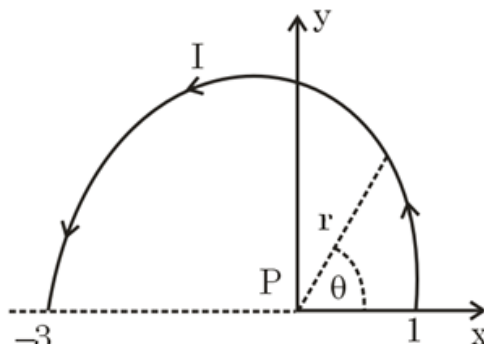
7. A thin metallic partition of negligible thickness is inserted between two shaded metallic plates as shown. The remaining ends are then packed with insulating plates to form a container like structure. 2 taps shown are opened at $t = 0$ and finally closed at $t = 5$ s. If the capacitance of system between A and B after closing taps is $x\epsilon_0$, Find the value of x . (Assume liquid to be non conducting) Volumetric flow rates and dielectric constants of liquid are given.



Space for Rough Work

8. A wire segment is bent into the shape of an Archimedes spiral (see figure). The equation that describes the curve in the range $0 \leq \theta \leq \pi$ is $r(\theta) = 1 + \frac{2}{\pi}\theta$, for $0 \leq \theta \leq \pi$, where θ is the angle from x-axis in radians.

Point P is at the origin. I is the current. The magnetic field strength at point P is B. The value of $\frac{\mu_0 I \log_e 3}{B}$ is :



9. An object A of mass 2 kg is moving on a frictionless horizontal track has perfectly inelastic collision with another object B of mass 3 kg made of the same material and moving in front of A in same direction. Their common speed after the collision is 4 m/s. Due to the collision the temperature of the two objects, which was initially the same, is increased, though only by 0.006°C . The specific heat capacities of the two objects are the same : $0.5 \text{ kJ/kg}^\circ\text{C}$. What was the initial speed (in m/s) of the colliding object A before the collision?

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
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Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

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Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. For a closed system involving only reversible work of expansion or compression, select the correct relationship

(A) $dU = dQ - PdV$

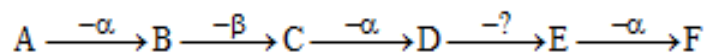
(B) $dH = dQ + VdP$

(C) $dH = dQ_p$

(D) none of these

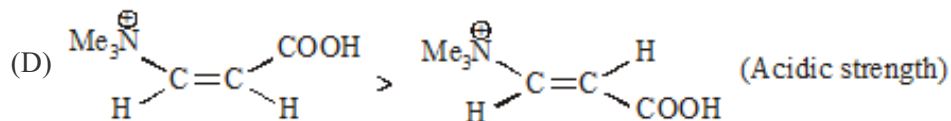
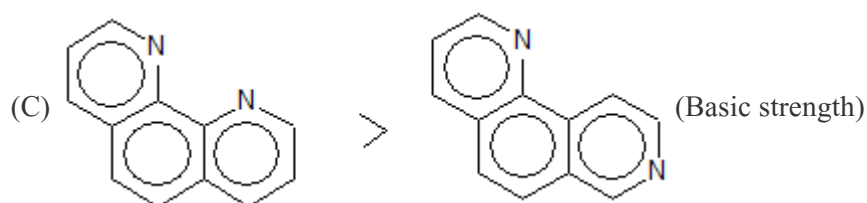
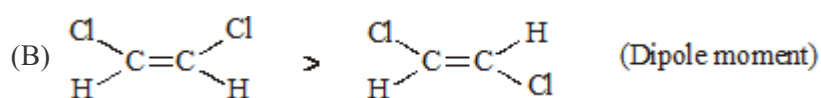
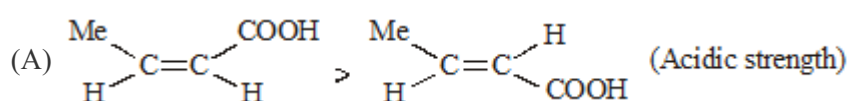
Space for Rough Work

2. For the following radioactive disintegration series.



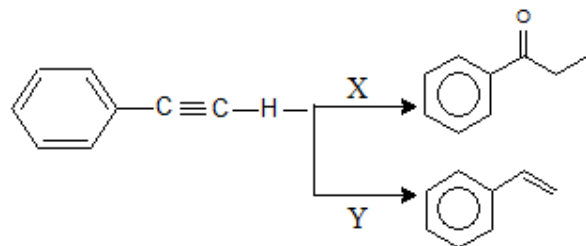
If it is known that B and E are isotopes and $E \xrightarrow{-\alpha} F$ means E shows α decay, then which of the options is/are correct?

- (A) D undergoes β^+ decay
 (B) The atomic number of A will be 89, if the atomic number of E is 87
 (C) The mass number of F will be 218, if the mass number of B is 226
 (D) The number of neutrons in C will be 138, If A has representation ${}_{89}A^{230}$
3. Which of the following is correct



Space for Rough Work

4. Consider following conversions



where X and Y are reagent or sets of reagents, which of the following statements is/are correct

- (A) Na/liq NH_3 can be used as Y
 (B) H_2/pd , BaSO_4 can be used as Y
 (C) $\text{NaNH}_2/\text{Me-I}$, 1% HgSO_4 , dil H_2SO_4 can be used as X
 (D) $\text{KOH}/\text{Me-I}$, 1% HgSO_4 , dil H_2SO_4 can be used as X
5. Which of the following informations is/are correct for the element with atomic number 50?
- (A) It is the member of d-Block
 (B) It belongs to the 5th period of periodic table
 (C) It has the minimum value of ionization energy in the group
 (D) It has 2 electrons in the valence shell
6. For the complex $\text{Na}_2 [\text{Fe} (\text{CN})_5 (\text{NO})]$ the true statements is/are
- (A) It is formed in the brown ring test of nitrate
 (B) It has 18 valence electrons on central metal
 (C) It's spin magnetic moment is zero
 (D) It produce violet/purple solution with Na_2S (aq.)

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Radius ratio: The arrangement of the ions in the crystal determined the relative size of cations and anions. One can calculate the limiting value of r_c/r_a known as radius ratio, from the packing of the ions. The critical ratio are

Cubic (8) $\underline{0.732}$ octahedral (6) $\underline{0.414}$ Tetrahedral (4) $\underline{0.225}$ Triangular (3) $\underline{0.155}$
Linear (2)

These ratios are approximate in number since in actual calculations, we must also consider the mutual repulsion and attraction of like and unlike charges respectively, as well as polarization of one ion by another

It is impossible for both anions and cations to have closest packed structures but if one of the ions is much larger than the other, it is common for the larger ions alone to approach a closest packed structure and smaller ions to fit into voids in this structure.

Answer the following questions:

7. For an ionic compound $M^+ X^-$, $\frac{r_{M^+}}{r_{X^-}} = 1.2$, Then what is the coordination no. of X^-
- (A) 4 (B) 6 (C) 8 (D) 3
8. For an ionic compound select the incorrect statment
- (P) Both cation and anion can form closest packed structure simultaneously
 (Q) It is common for the larger ions alone to approach a closest-packed structure.
 (R) In an ionic compound cation always goes into void
 (S) In an ionic compound cation and anion always in contact
- (A) P, Q (B) P, Q, R (C) P, R (D) Q, S

Space for Rough Work

Paragraph for Questions 9 and 10

Determination of ΔG , ΔS and ΔH for the cell reaction:

The following equations can be derived from Gibbs's Helmholtz equation and

$$(i) \Delta G = -nFE \quad (ii) \Delta G = \Delta H - nFT \left[\frac{\partial E}{\partial T} \right]_P$$

$$(iii) \Delta S = nF \left[\frac{\partial E}{\partial T} \right]_P \quad (iv) \Delta H = nF \left[T \left[\frac{\partial E}{\partial T} \right]_P - E \right]$$

($F = 96500 \text{ C}$)

Where $\left[\frac{\partial E}{\partial T} \right]_P$ = temperature coefficient of the emf of the cell

(i) $\left[\frac{\partial E}{\partial T} \right]_P = 0$, no heating or cooling of the cell occurs during its working

(ii) $\left[\frac{\partial E}{\partial T} \right]_P = +ve$, temperature of the cell falls during its working

(iii) $\left[\frac{\partial E}{\partial T} \right]_P = -ve$, temperature of the cell increases during its working

On the basis of above data, answer the following questions:

9. The e.m.f. of a standard Cd-cell is 1.018 V at 25°C. The temperature coefficient of the cell is $-5 \times 10^{-5} \text{ VK}^{-1}$. Calculate ΔS for the cell reaction {for $n = 2$ }.
- (A) $9.65 \text{ JK}^{-1} \text{ mol}^{-1}$ (B) $19.30 \text{ JK}^{-1} \text{ mol}^{-1}$
 (C) $-9.65 \text{ JK}^{-1} \text{ mol}^{-1}$ (D) $1.930 \text{ JK}^{-1} \text{ mol}^{-1}$
10. The temperature coefficient of a galvanic cell is $+5.0 \times 10^{-5} \text{ VK}^{-1}$. During the discharge of the cell, the cell temperature:
- (A) increase (B) decrease
 (C) does not change (D) first increases and then decrease

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A compound (A) of molecular formula $C_{14}H_{10}O_2$ is formed from $C_{14}H_{14}O_2$ by oxidation using $Cr_2O_7^{2-}$, solution

(A) upon treatment with OH^- followed by neutralization gives (B) which on treatment with conc. H_2SO_4/Δ gives (C) of molecular formula $C_{28}H_{20}O_4$ (B) responds to $NaHCO_3$ test and gives effervescence.

1. What should be the molecular wt of (B) is
2. Maximum no. of atom lying in same plane in compound (C) is

Space for Rough Work

Paragraph for Questions 3 and 4

Hydrogen peroxide an important chemical that can be used as an oxidising as well as reducing agent since the oxygen of this compound exists in -1 oxidation state

3. How many moles of H_2O_2 can be oxidised by 2 moles of KMnO_4 in acidic medium?
4. In the reaction $x\text{H}_2\text{O}_2 + \text{Cr}_2\text{O}_3 + \text{NaOH} \rightarrow \text{products}$, the stoichiometric coefficient x is?
(excess)

Paragraph for Questions 5 and 6

The complex compounds show space isomerism when the ligand species have different spatial arrangement surrounding the central metal.

5. What is the number of stereo isomeric forms for the complex $[\text{Pt}(\text{py})(\text{NH}_3)(\text{Br})(\text{I})]$
6. How many stereo isomers are possible for $[\text{Cr}(\text{H}_2\text{O})_2(\text{C}_2\text{O}_4)_2]^-$?

Space for Rough Work

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. How many of given methods are used to prepare different Lyophobic sols?
Peptization, exchange of solvent, electrical dispersion, mechanical dispersion, condensing the vapour
8. If molecular formula for pentose pentasaccharide is $C_aH_bO_c$, then the value of $a + b - c$ is?
9. How many of the following compounds give halogen acid (HX) as the product on hydrolysis at room temperature?
 BF_3 , BCl_3 , NCl_3 , PCl_3 , XeF_4 , SF_6 , ICl , BrF_5

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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. System of equation $x + y + az = b$, $2x + 3y = 2a$ & $3x + 4y + a^2z = ab + 2$ has

- | | |
|--|---|
| (A) unique solution when $a \neq 0$, $b \in \mathbb{R}$ | (B) no solution when $a = 0$, $b = 1$ |
| (C) infinite solution when $a = 0$, $b = 2$ | (D) infinite solution when $a = 1$, $b \in \mathbb{R}$ |
-

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2. Let two planes $P_1 : 2x - y + z - 2 = 0$ and $P_2 : x + 2y - z - 3 = 0$ are given then-
- (A) The equation of the plane through line of intersection of $P_1 = 0$ and $P_2 = 0$ and the point $(3, 2, 1)$ is $x - 3y + 2z + 1 = 0$
- (B) The equation of the plane through line of intersection of $P_1 = 0$ and $P_2 = 0$ and the point $(3, 2, 1)$ is $3x - y + 2z - 9 = 0$
- (C) The equation of acute angle bisector plane of $P_1 = 0$ and $P_2 = 0$ is $x - 3y + 2z + 1 = 0$
- (D) The equation of acute angle bisector plane of $P_1 = 0$ and $P_2 = 0$ is $x + 3y + 2z + 2 = 0$
3. Let $a_1, a_2, a_3 \in \mathbb{R}_0$ and $a_1 + a_2 \cos(2x) + a_3 \sin^2(x) = 0$ for all $x \in \mathbb{R}$, then
- (A) vectors $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ and $\vec{b} = 4\hat{i} + 2\hat{j} + \hat{k}$ are perpendicular to each other.
- (B) vectors $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ and $\vec{b} = -\hat{i} + \hat{j} + 2\hat{k}$ are parallel to each other.
- (C) If vector $\vec{a} = a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}$ is of length $\sqrt{6}$, then one of the ordered triplet (a_1, a_2, a_3) is equal to $(1, -1, -2)$.
- (D) If $2a_1 + 3a_2 + 6a_3 = 26$, then $|a_1 \hat{i} + a_2 \hat{j} + a_3 \hat{k}|$ is equal to $2\sqrt{6}$.

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4. Let S be the set of positive value(s) of r such that the area enclosed by the tangents drawn from the point P(6,8) to the circle $x^2 + y^2 = r^2$ and the chord of contact is $\frac{75\sqrt{3}}{4}$ sq. units. Then the correct option(s) is/are -
- (A) S is singleton set
- (B) Equation of chord of contact is $6x + 8y = \alpha^2$, $\alpha \in S$
- (C) Let A and B are point of contact of tangents then the equation of circumcircle of $\triangle ABP$ is $x^2 + y^2 - 6x - 8y = 0$
- (D) Number of elements in set S is 2

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5. Let z_1 & z_2 be points for which $|z_1|$ and $|z_2|$ are greatest and least values of z respectively lying on the circle $|z - 3 - 2i| = r$. If $t = z_1 + \bar{z}_1 + z_2 + \bar{z}_2$, then-

- (A) Maximum and minimum values of t depends on r
 (B) Maximum and minimum values of t is independent of r

(C) $\int_0^t \{x\} dx = 6$

(D) $\int_0^t \{x\} dx = 6r$

(where $\{ \}$ represent fractional part function)

6. If A and B are two orthogonal matrices of order 3, then -

- (A) A and B both will be invertible matrices
 (B) matrix ABA will also be orthogonal
 (C) matrix A^2B^2 will also be orthogonal
 (D) maximum value of $\det \left(\frac{A}{2} \text{adj}(2B) \right)$ is 8.

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SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A bag contains 4 white, 3 black, 2 red balls. Balls are drawn from the bag one by one without replacement.

7. The probability that 4th balls is red
- (A) $\frac{1}{16}$ (B) $\frac{1}{9}$ (C) $\frac{2}{9}$ (D) $\frac{3}{11}$
8. The probability that 5th ball is black given that 4th ball is red is equal to :
- (A) $\frac{1}{8}$ (B) $\frac{3}{8}$ (C) $\frac{2}{11}$ (D) $\frac{3}{11}$

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Paragraph for Questions 9 and 10

Consider a parabola $y^2 = 4x$.

9. Area of the figure formed by tangents and normals drawn at the extremities of its latus rectum is-
- (A) 8 (B) 16 (C) 4 (D) 2
10. The co-ordinates of centre of a circle circumscribing a triangle formed by any tangent, normal and x-axis is-
- (A) (0,0) (B) (1,0) (C) (1,1) (D) (0,1)

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SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider a tetrahedron PQRS with $PQ = 4$ units. A plane ' P_1 ' is drawn perpendicular to PQ and passing through P. Let P', R' & S' denotes the respective foot of perpendiculars drawn from P, R & S respectively to the plane ' P_1 ' such that area of $\Delta P'R'S' = 15$ sq. units.

1. Volume of the tetrahedron PQRS (in cu. units) is -
2. If the perpendicular distance of S' from plane PQR is 5 units then the perpendicular distance of R from line through PQ is -

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Paragraph for Questions 3 and 4

Consider four positive numbers a, b, c, d . Let first three are in a decreasing G.P and last three are in an A.P. The sum of extreme terms is 66 and sum of middle terms is 60.

Let U_n is n^{th} term of G.P $a, b, c, \dots$ & T_n is n^{th} term of A.P $b, c, d, \dots$, then

3. Sum of infinite terms of G.P whose first three terms are a, b, c respectively is-
4. $\sum_{i=1}^{\infty} U_i T_i$ is-

Paragraph for Questions 5 and 6

Let $f(x) = \frac{e^x}{x}$ and $g(x) = e^x$.

5. If $f(x) = k$ has two distinct roots, then minimum positive integral value of k is -
6. The number of integers in the interval in which $f(x) + g(x)$ is decreasing, is -

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SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. If z is not purely real then area between the curves $\text{Im}g \left(z + \frac{1}{z} \right) = 0$ & $|z - 1| = 2$ is $k\pi^2$ (in square units), then find k -
8. Given $I_1 = \int_0^1 (1 - x^{2021})^{\frac{1}{2022}} dx$ and $I_2 = \int_0^1 (1 - x^{2022})^{\frac{1}{2021}} dx$, then $\frac{3I_1 + 4I_2}{2I_1 - I_2}$ is equal to
9. Let x is the number of 6 digit even numbers such that sum of their digits is 50, then $\frac{x}{10}$ is equal to

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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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